Mitosisphere June Challenge

Exam Structure

This month's challenge will consist of 10 MCQ questions (1 pt each), 2 MTF questions (2 pts each), 2 FRQ questions (3 pts each), and 1 hard FRQ tiebreak. You will have 22 minutes to complete the exam.

Section 1: MCQ (10 Questions)

Anatomy and Physiology (2 Questions)

Question 1:
How disappointing! After spending 12 years of your life getting a PHD in biology you have returned to your high school to work as a biology teacher. During your lectures about the digestive system one of your students asks you about the difference between Cholecystokinin and Secretin. Recalling your USABO days you immediately state that:

a: Cholecystokinin is primarily involved in bile release, while Secretin is primarily involved in enzyme production

b: Cholecystokinin is primarily involved in bile release, while Secretin is primarily involved in enzyme release

c: Secretin is primarily involved in bile release, while Cholecystokinin is primarily involved in enzyme production

d: Secretin is primarily involved in bile production, while Cholecystokinin is primarily involved in enzyme release

e: Secretin is stimulated by amino acids and fats while Cholecystokinin is stimulated by low pH

Question 2:
You are investigating cell membranes in different organisms. Unfortunately, you accidentally ingested some of the shrooms before testing so none of the samples are labeled. However, it seems that you somehow took note of certain compounds found in large quantities in each of the 3 samples. Your data is given below:

Sample 1: Beta-sitosterol

Sample 2: Ergosterol

Sample 3: Cholesterol

Which of the following could possibly represent an organism for each of these samples?

a: Strawberry, Oyster mushroom, Bat

b: Shitaki mushroom, Blueberry, Camel

c: Human, Pineapple, Turtle

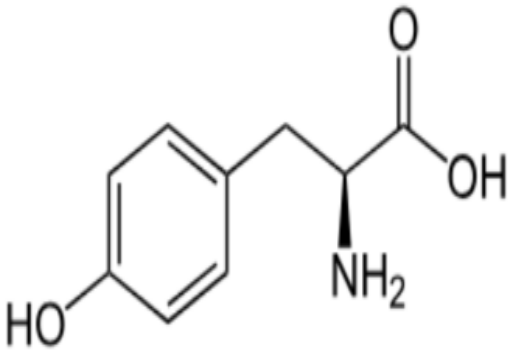
d: Trypanosoma, Agave, Fox

e: E.coli, Taro, Panda

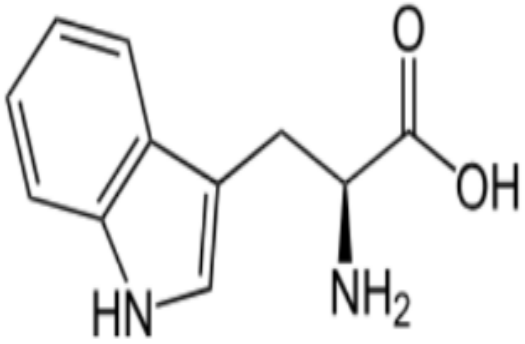
Cell Biology (2 Questions)

Question 3:
You are working at the Mitosisphere clinic when your patient reports symptoms of insomnia, anxiety, and impaired memory. Suspecting a hormonal deficiency, you prescribe which of the following hormone precursors?

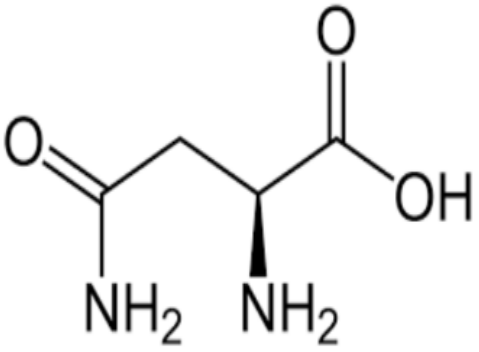
a.



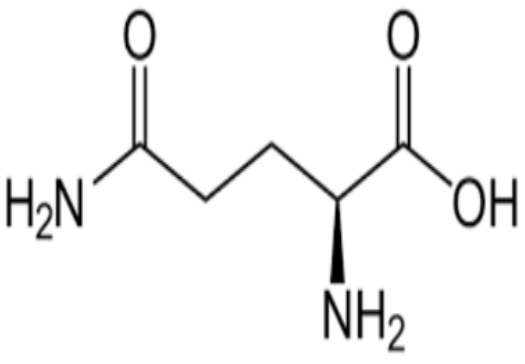
b.

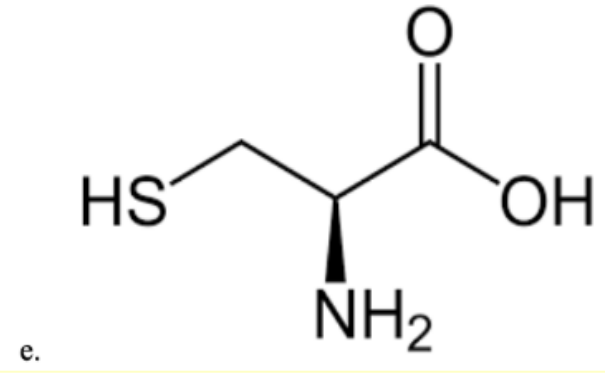


c.



d.





e.

Question 4:
Your lab is currently looking at the mechanisms in which deamination of certain nucleic acids affects their bonding properties. As part of your preemptive research your lab PI has you learn about the deamination of adenine. As a studied USABO student you immediately tell your PI that such research is unnecessary as you already know that:

a: Xanthine, pairs with adenine

b: Xanthine, pairs with cytosine

c: Xanthine, pairs with guanine

d: Hypoxanthine, pairs with cytosine

e: Hypoxanthine, pairs with guanine

Plant Biology (2 Questions)

Question 5:
Gibberellins (GAs) were first discovered in 1934 by T. Yabuta and Y. Sumiki during their research on a fungal disease affecting rice. Since then, over 100 different gibberellins have been identified, each playing vital roles in plant development. One key role of GAs is in seed germination, where they promote the production of hydrolytic enzymes. These enzymes break down the stored nutrients in the seed's endosperm, converting them into usable molecules like sugars and amino acids. In barley seeds, the molecules that are created by the broken down endosperm by such enzymes are quickly sent over to what part of the seed to be absorbed?

a: Pericarp

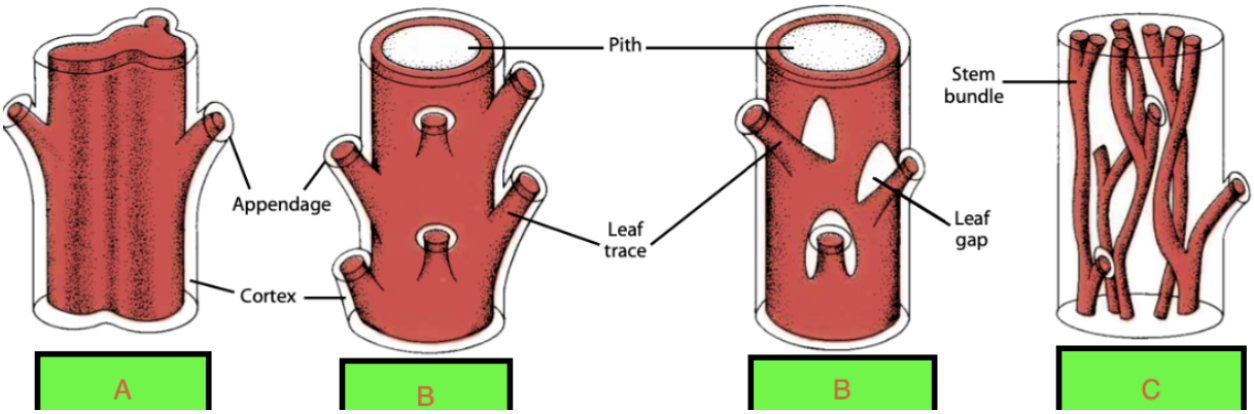
b: Aleurone layer

c: Scutellum

d: Testa

e: Coleoptile

Question 6:
Which of the following correctly describes the following type of steles?



a: Protostele, Sinophostele, Eustele

b: Sinophostele, Protostele, Eustele

c: Eustele, Sinophostele, Protostele

d: Protostele, Eustele, Sinophostele

e: Sinophostele, Eustele, Protostele

Genetics (1 Question)

Question 7:
Eric is researching the KINDBEFRIENDER operon, which encodes for the three polypeptides Nice, Caring, and Helpful. These three genes are flanked by an operator (KbO) and a promoter (KbP). He decides to study the activity of the KINDBEFRIENDER operon under various conditions as shown in the table below.

Cells exposed to the protein "*Mean*"

	Nice	Caring	Helpful
KbO+,KbP+	++	+	++
KbO+,KbP-	+	-	+
KbO-,KbP+	+	-	+
KbO-,KbP-	-	-	-

(A "++" indicates significant translation, a "+" indicates baseline level translation, a "-" represents insignificant translation)

Cells exposed to the protein "*Nice*"

	Nice	Caring	Helpful
KbO+,KbP+	-	+	-
KbO+,KbP-	-	-	-
KbO-,KbP+	+	-	+
KbO-,KbP-	-	-	-

Identify which of the following statements is NOT true.

a: The protein Mean acts as a positive regulator for transcription of the Helpful gene.

b: The KINDBEFRIENDER promoter increases expression of all three genes.

c: The protein Nice acts as a negative regulator for transcription of the Caring gene.

d: The protein Nice has a baseline level of expression under standard conditions.

e: The KINDBEFRIENDER operon is repressible.

Ecology (1 Question)

Question 8:
In a mark recapture experiment, Daniel labels 30 crocodiles red and releases them. After three days, he returns and is able to locate 50 crocodiles, however only 8 of them are labeled red. Which of the following is closest to the population size of these crocodiles?

a: 65 individuals

b: 190 individuals

c: 545 individuals

d: 1280 individuals

e: 12000 individuals

Ethology (1 Question)

Question 9:
You are watching anime when suddenly you question the real life applicability of the word harem within nature. Which of the following relationships best describes a harem of multiple females with a single male?

a: Polyandry

b: Polygyny

c: Polymorphy

d: Monogamy

e: Promiscuity

Biosystematics (1 Question)

Question 10:
Ryan the red rhino has unfortunately lost his parents and is looking for a new home. However, he is the last member of his species so you decide to instead find the closest related animal to foster him. Which of the following animals would be the best choice?

a: Harry the horse

b: Greg the gorilla

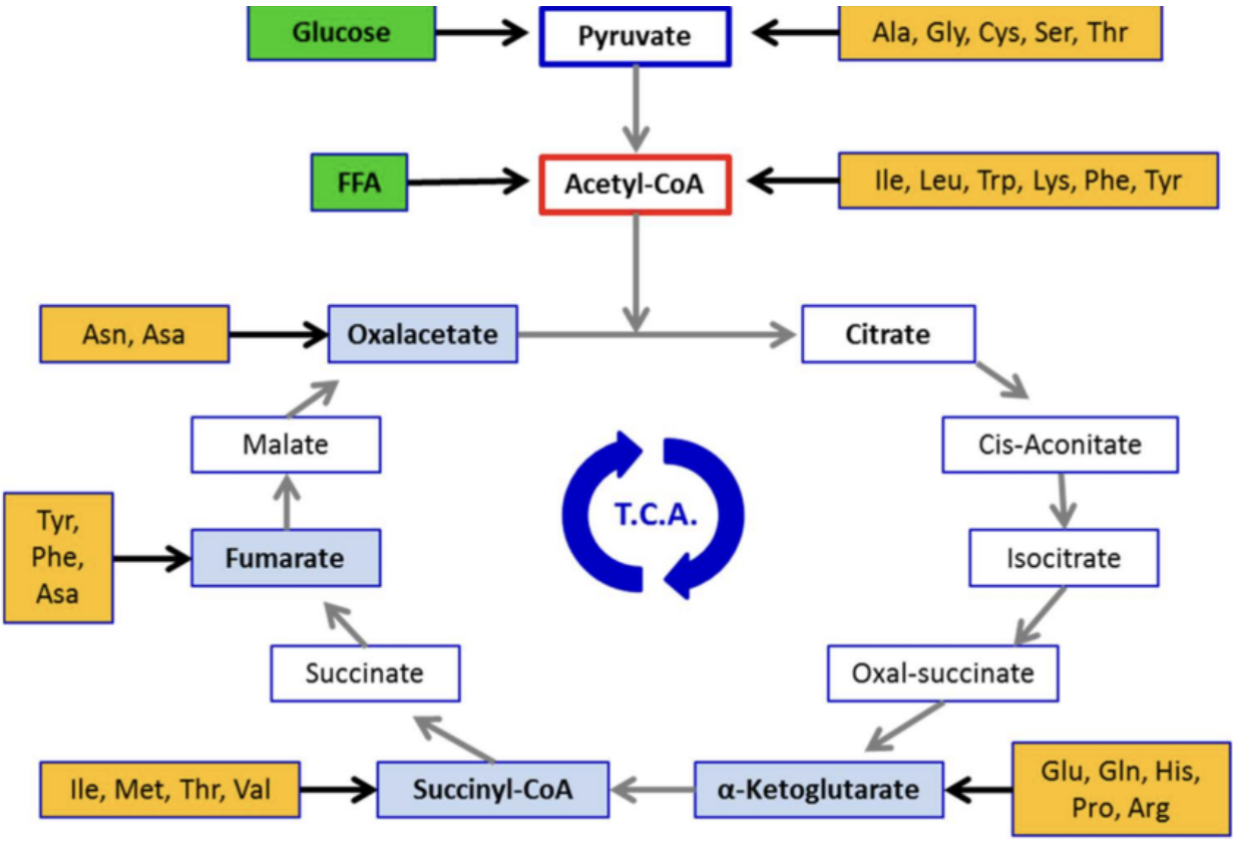
c: Sandy the sandshrew

d: Florence the fox

e: Curie the camel

Section 2: MTF (2 Questions)

Question 11:
Using the image below, determine whether the following statements about the krebs cycle are true or false.



a: Valine to Krebs intermediate yields 1 FADH2, 1 NADH, 1 ATP

b: Tyrosine to Krebs intermediate yields 1 FADH2, 3 NADH, 1 ATP

c: Succinyl-CoA to succinate is irreversible

d: Methionine is the only AA from Acetyl-CoA

Source: Pasini, Evasio & Corsetti, Giovanni & Aquilani, Roberto & Romano, Claudia & Picca, Anna & Calvani, Riccardo & Dioguardi, Francesco. (2018). Protein-Amino Acid Metabolism Disarrangements: The Hidden Enemy of Chronic Age-Related Conditions. *Nutrients.* 10, 391. https://doi.org/10.3390/nu10040391

Question 12:
Would the following situations result(true) or not result(false) in an influx of ions for a hypothetical neuron cell with only voltage gated magnesium and chloride channels at a membrane equilibrium potential of -60 mv? Use the nerst equation as given: 61/x(charge) * log(Concentration of ion outside/ concentration of ion inside)

a: Mg^{2+}: 2 mM outside, 150 mM inside

b: Mg^{2+}: 0.5 mM inside, 50 mM inside (typo — clarify?)

c: Cl^{-}: 2 mM outside, 10 mM inside

d: Cl^{-}: 3 mM outside, 35 mM inside

Question 13: Lost in Translation

The Mitosisphere research team is working on creating a transgenic bacterium engineered to express the human alpha hemoglobin gene. You transduce a bacterial cell with a `pYES2` yeast plasmid vector and await your results!

(a) **You have decided to include a special amino acid outside of the normal 20 in your protein sequence. Which amino acid could this be? Compared to other amino acids, what is special about the way this amino acid is incorporated?**

(b) **After painstakingly figuring out all issues with transcription and gene expression, the protein product is still dysfunctional! Identify *two* possible reasons this could be.**

(c) **Propose a strategy to help alleviate *one* of these issues.**

Question 14: Powder Pandemonium

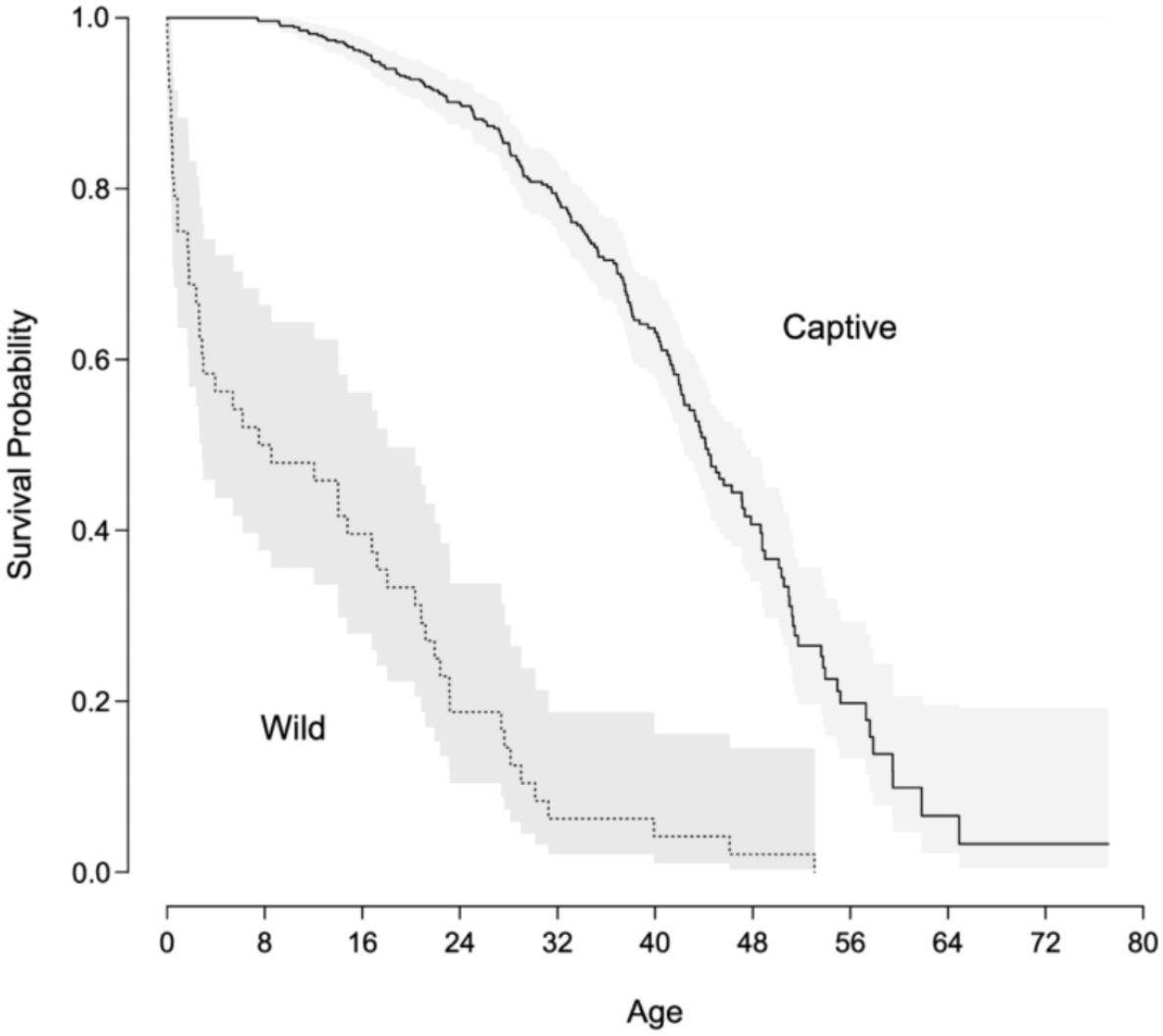
Oh no! Your opponents' powder monkeys are rapidly growing and threaten to end the game. Strangely enough, the growth of the powder monkey population follows a logistic growth curve. Each turn you record data regarding the number of powder monkeys, which is shown below:

Turn	# of Monkeys
0	10
1	15
2	20
3	26
4	32
5	38
6	42
7	45
8	47
9	48
10	49
11	49
12	50
13	50
14	50
15	50

(a) **What is the carrying capacity of this population? In a real-life scenario, is it possible to pass this threshold?**

(b) **Given that the intrinsic rate of growth (r) is 0.5, derive a logistic equation to represent the number of monkeys, N, at a given time t.**

(c) **You decide to test the lifespan of powder monkeys within capacity before they ultimately meet their explosive doom. What type of survivorship curve best describes the captive and wild powder monkeys respectively using the data below?**



Tiebreaker!

Metazoa: Identify what animal family the following images are from. This question will only be used in the case of a tiebreak.

A.



B.



C. 

D. 

E. 

Answer bank: Accipitridae, Anatidae, Aplysiidae, Bovidae, Bufonidae, Canidae, Carcharhinidae, Cervidae, Cheloniidae, Colubridae, Crocodylidae, Dasyatidae, Delphinidae, Equidae, Felidae, Formicidae, Gekkonidae, Hirudinidae, Hominidae, Hyaenidae, Ichneumonidae, Leporidae, Loricariidae, Macropodidae, Muridae, Octopodidae, Papilionidae, Phasianidae, Psittacidae, Ranidae, Scarabaeidae, Sciuridae, Spheniscidae, Sirenidae, Tenrecidae, Testudidae, Tyrannidae, Ursidae, Vespidae, Vulpesidae.

Source: Wikipedia and Altschul et al. (2018) eLife 7: e33781.